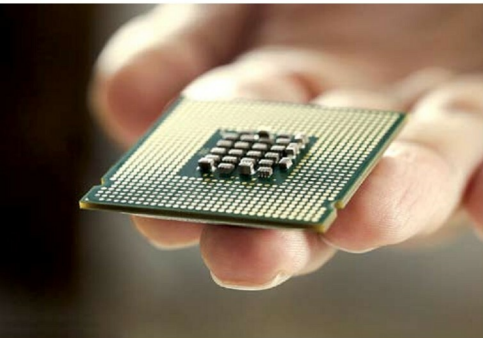


EMBEDDED SYSTEMS

Can India Move from Implementing to Initiating Products?

Currently, as most users are outside India, they decide what is needed next. In future, though, India can migrate from the role of product implementer to product initiator by focusing on the growing domestic market



and making it production-ready in all respects for commercial success.

The embedded ecosystem

The players in the embedded space include system integrators, IC design houses, electronic design automation vendors, board design houses, test houses, software tool providers, and embedded software and application software providers.

Most of the semiconductor manufacturers are involved in VLSI chip design and embedded software development. Indian companies are developing embedded products in consumer, industrial, telecom, automotive, computing and medical space for local as well as export markets.

Transnational global companies like Cisco, Juniper, Honeywell, GE, Bosch, Delphi, Visteon, Continental and many more have set up their embedded development centres in India. In addition, design services companies of India like Wipro, HCL, Satyam, Mindtree, L&T, TCS, Tata Elxsi, KPIT Cummins and many more are doing embedded software and hardware product development in India for global accounts.

India's forte: embedded software

The show stealer so far has, of course, been embedded software development. "Embedded software develop-

■ UMA BANSAL

Consumer electronics and telecom companies are starting to recognise the importance of localisation of their products, with India emerging as a big market for these products. Given this, the Indian embedded solution providers are in a unique position where they can add value not only in the development process but also in the

conceptualisation of new and unique products.

Currently, most of our embedded activities pertain to products developed for foreign markets. And most Indian embedded system houses are engaged in sub-system architecture, hardware and software implementation, testing and integration stages of the value chain. India can move up the value chain by conceiving and specifying products (as against service), system architecting, system validation